

An Information-Theoretic Perspective on Covert Uplink Communication in FD-ISAC Systems

System Model. We consider a simplified single-antenna full-duplex (FD) system that isolates the fundamental interaction between an uplink covert transmission and a concurrent downlink waveform. Alice communicates covertly with an FD base station (BS), while a passive warden Willie attempts to determine whether Alice is active. During a block of n channel uses, the BS simultaneously transmits an independent Gaussian downlink waveform

$$S_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, P_d), \quad i = 1, \dots, n, \quad (1)$$

where P_d is fixed. All channel coefficients are assumed constant over the block and known. Willie knows the distribution and power of S^n , but not its realization.

After self-interference cancellation, the BS observes

$$Y_i = h_{ab}X_i + \tilde{N}_{b,i}, \quad \tilde{N}_{b,i} \sim \mathcal{CN}(0, N_b(P_d)), \quad (2)$$

where $N_b(P_d) \triangleq \sigma_b^2 + \beta|h_{\text{SI}}|^2 P_d$ models receiver noise and residual self-interference. Here h_{ab} is the Alice–BS channel, h_{SI} is the self-interference channel, and $\beta \geq 0$ denotes the residual self-interference factor.

Willie’s observation is

$$Z_i = g_{aw}X_i + g_{bw}S_i + N_{w,i}, \quad N_{w,i} \sim \mathcal{CN}(0, \sigma_w^2), \quad (3)$$

where g_{aw} and g_{bw} denote the Alice–Willie and BS–Willie channels, respectively. Defining

$$V_i \triangleq g_{bw}S_i + N_{w,i} \sim \mathcal{CN}(0, N_w(P_d)), \quad N_w(P_d) \triangleq \sigma_w^2 + |g_{bw}|^2 P_d, \quad (4)$$

the surveillance channel becomes $Z_i = g_{aw}X_i + V_i$.

Code Definition. For blocklength n , Alice employs a fixed (n, M_n) codebook

$$\mathcal{C}_n = \{x^n(1), \dots, x^n(M_n)\} \subset \mathbb{C}^n, \quad (5)$$

with a uniformly distributed message $M \sim \text{Unif}\{1, \dots, M_n\}$. The encoder maps $M = m$ to the deterministic codeword $x^n(m)$, subject to the per-codeword power constraint

$$\frac{1}{n} \|x^n(m)\|^2 \leq P_a, \quad \forall m. \quad (6)$$

The BS decoder $\phi_n : \mathbb{C}^n \rightarrow \{1, \dots, M_n\}$ produces $\hat{M} = \phi_n(Y^n)$, and reliability requires

$$P_e^{(n)} \triangleq \Pr(\hat{M} \neq M) \rightarrow 0. \quad (7)$$

Willie’s Hypothesis Test and Coverttness. Under Alice’s silence,

$$\mathcal{H}_0 : \quad Z^n = V^n, \quad P_0^{(n)} = \mathcal{CN}(\mathbf{0}, N_w(P_d)\mathbf{I}_n). \quad (8)$$

Under $\mathcal{H}_1$, conditioned on message $M = m$ and the fixed codebook,

$$Z^n = g_{aw}x^n(m) + V^n, \quad (9)$$

and therefore

$$P_{Z^n|M=m, \mathcal{H}_1, \mathcal{C}_n} = \mathcal{CN}(g_{aw}x^n(m), N_w(P_d)\mathbf{I}_n). \quad (10)$$

Since Willie does not know M , the actual distribution induced by the fixed codebook is the Gaussian mixture

$$P_{1, \mathcal{C}_n}^{(n)} = \frac{1}{M_n} \sum_{m=1}^{M_n} \mathcal{CN}(g_{aw}x^n(m), N_w(P_d)\mathbf{I}_n). \quad (11)$$

We impose the coverttness constraint

$$D(P_{1, \mathcal{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta, \quad (12)$$

for a prescribed $\delta > 0$. The fundamental objective is to characterize the maximum codebook size

$$M_n^*(\delta) = \max_{\mathcal{C}_n, \phi_n} \left\{ M_n : P_e^{(n)} \rightarrow 0, D(P_{1, \mathcal{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta \right\}, \quad (13)$$

and, in particular, the asymptotic growth of $\log M_n^*(\delta)$ as $n \rightarrow \infty$ and its dependence on the concurrent downlink power P_d .

Theorem 1 (Achievable Square-Root Law for FD-ISAC). Assume $g_{aw} \neq 0$ and $P_a > 0$. If

$$\frac{|h_{ab}|^2}{N_b(P_d)} > \frac{|g_{aw}|^2}{N_w(P_d)}, \quad (14)$$

then, for every $\delta > 0$, there exists a sequence of deterministic codebooks $\{\mathcal{C}_n\}$ satisfying the per-codeword power constraint, such that the decoding error probability vanishes and the KL-divergence covertness constraint at Willie is satisfied, i.e.,

$$P_e^{(n)} \rightarrow 0, \quad D(P_{1,\mathcal{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta$$

for all sufficiently large n . Moreover, the maximum number of reliably and covertly transmissible messages $M_n^*(\delta)$ satisfies

$$\liminf_{n \rightarrow \infty} \frac{\log M_n^*(\delta)}{\sqrt{n}} \geq \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}. \quad (15)$$

Physical Interpretation. The achievable square-root coefficient depends on the concurrent downlink power P_d through

$$\frac{N_w(P_d)}{N_b(P_d)} = \frac{\sigma_w^2 + |g_{bw}|^2 P_d}{\sigma_b^2 + \beta |h_{SI}|^2 P_d}. \quad (16)$$

Increasing P_d therefore has two competing effects: it increases the uncertainty at Willie through the BS–Willie link, while simultaneously increasing the residual self-interference at the BS. In particular, the achievable coefficient increases with P_d if

$$|g_{bw}|^2 \sigma_b^2 > \beta |h_{SI}|^2 \sigma_w^2, \quad (17)$$

decreases with P_d when the inequality is reversed, and is independent of P_d at the square-root order when equality holds. The downlink power also affects the feasibility condition in Theorem 1 through the relative effective qualities of the Alice–BS and Alice–Willie channels.

Proof. The proof first selects the Gaussian signaling power to meet the covertness requirement at Willie. It then shows that the codebook can be chosen large enough to approximate the desired Willie-side distribution while remaining small enough for reliable decoding at the BS. The condition in Theorem 1 guarantees that these two requirements are compatible. A deterministic codebook satisfying reliability, the power constraint, and actual covertness is then extracted from the random-coding ensemble.

Random-Coding Construction. We begin with a Gaussian random-code ensemble. This choice induces a Gaussian output law at Willie that can be characterized explicitly and serves as the reference distribution for the subsequent covertness analysis. For each blocklength n , let $Q_n > 0$ and $M_n \in \mathbb{N}$. Generate the random codebook

$$\mathbf{C}_n = \{X^n(1), \dots, X^n(M_n)\}, \quad X_i(m) \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, Q_n), \quad (18)$$

independently over $i = 1, \dots, n$ and $m = 1, \dots, M_n$. A realization of $\mathbf{C}_n$ is denoted by

$$\mathcal{C}_n = \{x^n(1), \dots, x^n(M_n)\}.$$

For $M \sim \text{Unif}\{1, \dots, M_n\}$, Alice transmits $x^n(M)$. The BS employs the ML decoder

$$\phi_n(y^n) = \arg \min_{1 \leq m \leq M_n} \|y^n - h_{ab} x^n(m)\|^2. \quad (19)$$

To characterize the output distribution induced by the Gaussian codeword ensemble at Willie, recall that

$$Z_i = g_{aw} X_i + V_i, \quad V_i \sim \mathcal{CN}(0, N_w(P_d)).$$

Since X_i and V_i are independent Gaussian random variables,

$$Z_i \sim \mathcal{CN}(0, N_w(P_d) + |g_{aw}|^2 Q_n). \quad (20)$$

Independence across channel uses therefore gives the Gaussian reference distribution

$$\bar{P}_{1,n}^{(n)} \triangleq \mathcal{CN}(\mathbf{0}, [N_w(P_d) + |g_{aw}|^2 Q_n] \mathbf{I}_n). \quad (21)$$

Reference Covertness and Power Scaling. We first determine how the signaling power Q_n must scale with the blocklength n to satisfy the covertness constraint at Willie. The divergence between the Gaussian reference distribution in (17) and the innocent distribution $P_0^{(n)}$ in (8) are evaluated first. For zero-mean proper complex Gaussian distributions,

$$D(\mathcal{CN}(\mathbf{0}, \mathbf{\Sigma}_1) \| \mathcal{CN}(\mathbf{0}, \mathbf{\Sigma}_0)) = \log \frac{\det \mathbf{\Sigma}_0}{\det \mathbf{\Sigma}_1} + \text{tr}(\mathbf{\Sigma}_0^{-1} \mathbf{\Sigma}_1) - n.$$

Hence,

$$D(\bar{P}_{1,n}^{(n)} \| P_0^{(n)}) = n \left[\frac{|g_{aw}|^2 Q_n}{N_w(P_d)} - \log \left(1 + \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} \right) \right]. \quad (22)$$

Imposing

$$D(\bar{P}_{1,n}^{(n)} \| P_0^{(n)}) \leq \delta \quad (23)$$

gives

$$0 \leq \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} - \log \left(1 + \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} \right) \leq \frac{\delta}{n}. \quad (24)$$

Since $x - \log(1+x)$ is strictly increasing for $x \geq 0$ and vanishes only at $x = 0$, it follows that

$$Q_n \rightarrow 0. \quad (25)$$

The expansion

$$x - \log(1+x) = \frac{x^2}{2} + O(x^3), \quad x \rightarrow 0,$$

therefore yields

$$D(\bar{P}_{1,n}^{(n)} \| P_0^{(n)}) = \frac{n|g_{aw}|^4 Q_n^2}{2N_w^2(P_d)} (1 + o(1)). \quad (26)$$

Consequently,

$$Q_n \leq \frac{N_w(P_d)}{|g_{aw}|^2} \sqrt{\frac{2\delta}{n}} (1 + o(1)), \quad (27)$$

and hence

$$Q_n = O(n^{-1/2}). \quad (28)$$

Power-Sequence Selection. We now choose a specific signaling-power sequence consistent with the scaling derived above. Fix an arbitrary $0 < \varepsilon < 1$. Motivated by (23), choose

$$Q_n = (1 - \varepsilon) \frac{N_w(P_d)}{|g_{aw}|^2} \sqrt{\frac{2\delta}{n}}. \quad (29)$$

Substituting this choice into (22) gives

$$\begin{aligned} D(\bar{P}_{1,n}^{(n)} \| P_0^{(n)}) &= \frac{n|g_{aw}|^4 Q_n^2}{2N_w^2(P_d)} (1 + o(1)) \\ &= (1 - \varepsilon)^2 \delta (1 + o(1)) < \delta \end{aligned} \quad (30)$$

for all sufficiently large n . Moreover,

$$\sqrt{n} Q_n \rightarrow (1 - \varepsilon) \frac{N_w(P_d)}{|g_{aw}|^2} \sqrt{2\delta}, \quad (31)$$

and hence

$$Q_n = \Theta(n^{-1/2}). \quad (32)$$

Reliability at the BS. We next derive a sufficient upper-side condition on $\log M_n$ under which the BS can decode reliably. Under the Gaussian input $X_i \sim \mathcal{CN}(0, Q_n)$, Bob's channel in (2) induces

$$Y_i \sim \mathcal{CN}(0, N_b(P_d) + |h_{ab}|^2 Q_n).$$

Let $p_{Y|X}(\cdot|x)$ denote the conditional Gaussian density in (2), and let $p_{Y,n}$ denote the above output density. Define the single-letter information density by

$$i_{b,n}(x; y) \triangleq \log \frac{p_{Y|X}(y|x)}{p_{Y,n}(y)}, \quad (33)$$

and, by memorylessness,

$$i_{b,n}^{(n)}(X^n; Y^n) = \sum_{i=1}^n i_{b,n}(X_i; Y_i). \quad (34)$$

Its mean is the Gaussian-input mutual information

$$\begin{aligned} I_{b,n} &\triangleq \mathbb{E}[i_{b,n}(X_i; Y_i)] \\ &= \log \left(1 + \frac{|h_{ab}|^2 Q_n}{N_b(P_d)} \right), \end{aligned} \quad (35)$$

while direct evaluation of the Gaussian information density gives

$$\mathbb{E}[i_{b,n}^{(n)}(X^n; Y^n)] = nI_{b,n}, \quad \text{Var}[i_{b,n}^{(n)}(X^n; Y^n)] = \frac{2n|h_{ab}|^2 Q_n}{N_b(P_d) + |h_{ab}|^2 Q_n}. \quad (36)$$

Since the ML decoder minimizes the average decoding error for a fixed codebook, by Feinstein's random-coding bound, for any $\tau_n > 0$,

$$\mathbb{E}_{C_n}[P_e^{(n)}] \leq \Pr[i_{b,n}^{(n)}(X^n; Y^n) \leq \log M_n + \tau_n] + e^{-\tau_n}. \quad (37)$$

Define the reliability margin

$$\Delta_{b,n} \triangleq nI_{b,n} - \log M_n. \quad (38)$$

For $\Delta_{b,n} > 0$, choosing $\tau_n = \Delta_{b,n}/2$ and applying Chebyshev's inequality gives

$$\begin{aligned} \mathbb{E}_{C_n}[P_e^{(n)}] &\leq \frac{4 \text{Var}[i_{b,n}^{(n)}]}{\Delta_{b,n}^2} + e^{-\Delta_{b,n}/2} \\ &= \frac{8n|h_{ab}|^2 Q_n}{[N_b(P_d) + |h_{ab}|^2 Q_n] \Delta_{b,n}^2} + e^{-\Delta_{b,n}/2}. \end{aligned} \quad (39)$$

Since $Q_n = \Theta(n^{-1/2})$, the variance term is $\Theta(\sqrt{n})$. Hence

$$\frac{\Delta_{b,n}}{n^{1/4}} \rightarrow \infty \quad (40)$$

is sufficient to ensure

$$\mathbb{E}_{C_n}[P_e^{(n)}] \rightarrow 0. \quad (41)$$

Moreover,

$$\begin{aligned}
nI_{b,n} &= n \log \left(1 + \frac{|h_{ab}|^2 Q_n}{N_b(P_d)} \right) \\
&= \frac{|h_{ab}|^2}{N_b(P_d)} nQ_n + O(nQ_n^2) \\
&= (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + O(1).
\end{aligned} \tag{42}$$

Resolvability at Willie. We next study whether a finite random codebook can reproduce the Gaussian reference distribution at Willie. This will determine how large the codebook needs to be. Let $p_{Z|X}(z|x)$ denote the conditional density of Willie's channel,

$$p_{Z|X}(z|x) = \frac{1}{\pi N_w(P_d)} \exp \left(-\frac{|z - g_{aw}x|^2}{N_w(P_d)} \right), \tag{43}$$

and let $\bar{p}_{1,n}(z)$ denote the single-letter density associated with the Gaussian reference distribution in (17),

$$\bar{p}_{1,n}(z) = \frac{1}{\pi[N_w(P_d) + |g_{aw}|^2 Q_n]} \exp \left(-\frac{|z|^2}{N_w(P_d) + |g_{aw}|^2 Q_n} \right). \tag{44}$$

By construction,

$$\bar{p}_{1,n}(z) = \mathbb{E}_X [p_{Z|X}(z|X)], \quad X \sim \mathcal{CN}(0, Q_n). \tag{45}$$

Define Willie's single-letter and n -letter information densities as

$$\imath_{w,n}(x; z) \triangleq \log \frac{p_{Z|X}(z|x)}{\bar{p}_{1,n}(z)}, \tag{46}$$

and

$$\imath_{w,n}^{(n)}(x^n; z^n) \triangleq \sum_{i=1}^n \imath_{w,n}(x_i; z_i). \tag{47}$$

The density of the reference distribution is therefore

$$\bar{p}_{1,n}^{(n)}(z^n) = \prod_{i=1}^n \bar{p}_{1,n}(z_i),$$

and the random codebook $\mathcal{C}_n$ induces

$$p_{1,\mathcal{C}_n}^{(n)}(z^n) = \frac{1}{M_n} \sum_{m=1}^{M_n} \prod_{i=1}^n p_{Z|X}(z_i | X_i(m)). \tag{48}$$

Hence,

$$\frac{p_{1,\mathcal{C}_n}^{(n)}(z^n)}{\bar{p}_{1,n}^{(n)}(z^n)} = \frac{1}{M_n} \sum_{m=1}^{M_n} \exp \left[\imath_{w,n}^{(n)}(X^n(m); z^n) \right]. \tag{49}$$

By symmetry of the i.i.d. random codebook, let $X^n(1)$ denote the transmitted codeword and generate Z^n according to $\prod_{i=1}^n p_{Z|X}(z_i | X_i(1))$. Then

$$\begin{aligned}
&\mathbb{E}_{\mathcal{C}_n} D \left(P_{1,\mathcal{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)} \right) \\
&= \mathbb{E} \log \left[\frac{1}{M_n} \sum_{m=1}^{M_n} \exp \left(\imath_{w,n}^{(n)}(X^n(m); Z^n) \right) \right].
\end{aligned} \tag{50}$$

For every $j \neq 1$, $X^n(j)$ is independent of Z^n , and

$$\begin{aligned}
\mathbb{E}_{X^n(j)} \exp \left[\imath_{w,n}^{(n)}(X^n(j); z^n) \right] &= \prod_{i=1}^n \frac{\mathbb{E}_X [p_{Z|X}(z_i | X_i)]}{\bar{p}_{1,n}(z_i)} \\
&= 1.
\end{aligned} \tag{51}$$

Conditioning on $(X^n(1), Z^n)$ and applying Jensen's inequality to $X^n(2), \dots, X^n(M_n)$ therefore gives

$$\begin{aligned}
\mathbb{E}_{\mathcal{C}_n} D \left(P_{1,\mathcal{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)} \right) &\leq \mathbb{E} \log \left[\frac{e^{\imath_{w,n}^{(n)}(X^n(1); Z^n)} + M_n - 1}{M_n} \right] \\
&= \mathbb{E} \log \left[1 + \frac{e^{\imath_{w,n}^{(n)}(X^n(1); Z^n)} - 1}{M_n} \right] \\
&\leq \mathbb{E} \log \left[1 + \frac{e^{\imath_{w,n}^{(n)}(X^n(1); Z^n)}}{M_n} \right].
\end{aligned} \tag{52}$$

For any fixed $0 < \alpha < 1$, using $\log(1+t) \leq t^\alpha/\alpha$ for $t \geq 0$ yields

$$\begin{aligned}
\mathbb{E}_{\mathcal{C}_n} D \left(P_{1,\mathcal{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)} \right) &\leq \frac{1}{\alpha M_n^\alpha} \mathbb{E} \exp \left[\alpha \imath_{w,n}^{(n)}(X^n(1); Z^n) \right] \\
&= \frac{1}{\alpha M_n^\alpha} \left(\mathbb{E} e^{\alpha \imath_{w,n}^{(n)}(X; Z)} \right)^n,
\end{aligned} \tag{53}$$

where the last equality follows from the memoryless channel and the i.i.d. Gaussian input.

Gaussian Exponential Moment at Willie. To make the preceding one-shot resolvability bound explicit, it remains to evaluate the exponential moment of Willie's information density under the Gaussian input law. Define

$$s_{w,n} \triangleq \frac{|g_{aw}|^2 Q_n}{N_w(P_d)}. \tag{54}$$

Under the joint distribution of (X, Z) , define $U \triangleq \frac{g_{aw}X}{|g_{aw}|\sqrt{Q_n}}$, and $W \triangleq \frac{Z - g_{aw}X}{\sqrt{N_w(P_d)}}$. Then

$$U, W \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, 1), \quad \frac{Z}{\sqrt{N_w(P_d)}} = \sqrt{s_{w,n}} U + W.$$

Substitution into (42) gives

$$i_{w,n}(X; Z) = \log(1 + s_{w,n}) - |W|^2 + \frac{|\sqrt{s_{w,n}} U + W|^2}{1 + s_{w,n}}. \quad (55)$$

Let $\boldsymbol{\xi} \triangleq \begin{bmatrix} U \\ W \end{bmatrix} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_2)$. Then

$$i_{w,n}(X; Z) = \log(1 + s_{w,n}) + \boldsymbol{\xi}^H \mathbf{A}_{w,n} \boldsymbol{\xi}, \quad (56)$$

where

$$\mathbf{A}_{w,n} = \frac{1}{1 + s_{w,n}} \begin{bmatrix} s_{w,n} & \sqrt{s_{w,n}} \\ \sqrt{s_{w,n}} & -s_{w,n} \end{bmatrix}. \quad (57)$$

For $0 < \alpha < 1$, $\mathbf{I}_2 - \alpha \mathbf{A}_{w,n} \succ 0$, since $1 - \frac{\alpha s_{w,n}}{1 + s_{w,n}} > 0$ and

$$\det(\mathbf{I}_2 - \alpha \mathbf{A}_{w,n}) = 1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}} > 0. \quad (58)$$

Hence the proper-complex Gaussian integral gives

$$\begin{aligned} \mathbb{E} e^{\alpha i_{w,n}(X; Z)} &= (1 + s_{w,n})^\alpha \mathbb{E} e^{\alpha \boldsymbol{\xi}^H \mathbf{A}_{w,n} \boldsymbol{\xi}} \\ &= \frac{(1 + s_{w,n})^\alpha}{\det(\mathbf{I}_2 - \alpha \mathbf{A}_{w,n})} \\ &= \frac{(1 + s_{w,n})^\alpha}{1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}}}. \end{aligned} \quad (59)$$

Substituting into (49),

$$\mathbb{E}_{\mathbf{C}_n} D(P_{1, \mathbf{C}_n}^{(n)} \parallel \bar{P}_{1, n}^{(n)}) \leq \frac{1}{\alpha M_n^\alpha} \left[\frac{(1 + s_{w,n})^\alpha}{1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}}} \right]^n. \quad (60)$$

Asymptotic Resolvability. We now show that the resolvability bound vanishes when the codebook is chosen sufficiently large. Define Willie's Gaussian-input mutual information as

$$I_{w,n} \triangleq \log(1 + s_{w,n}). \quad (61)$$

The preceding resolvability bound can be written, for any $0 < \alpha < 1$, as

$$\begin{aligned} \mathbb{E}_{\mathbf{C}_n} D(P_{1, \mathbf{C}_n}^{(n)} \parallel \bar{P}_{1, n}^{(n)}) \\ \leq \exp \left\{ -\log \alpha - \alpha(\log M_n - n I_{w,n}) - n \log \left(1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}} \right) \right\}. \end{aligned} \quad (62)$$

From the selected power sequence,

$$s_{w,n} = \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} = (1 - \varepsilon) \sqrt{\frac{2\delta}{n}}, \quad (63)$$

so that

$$s_{w,n} = \Theta(n^{-1/2}), \quad n s_{w,n} = \Theta(\sqrt{n}). \quad (64)$$

Consider codebook sizes satisfying

$$\log M_n - n I_{w,n} \geq n^{3/8}. \quad (65)$$

Since the one-shot bound holds for every $0 < \alpha < 1$ at each n , choose

$$\alpha_n \triangleq \frac{n^{3/8}}{4n s_{w,n}}. \quad (66)$$

Then

$$\alpha_n = \Theta(n^{-1/8}) \longrightarrow 0, \quad (67)$$

and hence $0 < \alpha_n < 1$ for all sufficiently large n . Moreover,

$$\frac{\alpha_n^2 s_{w,n}}{1 + s_{w,n}} \longrightarrow 0.$$

Using $-\log(1 - u) \leq 2u$ for sufficiently small $u \geq 0$ gives

$$\begin{aligned} \mathbb{E}_{\mathbf{C}_n} D(P_{1, \mathbf{C}_n}^{(n)} \parallel \bar{P}_{1, n}^{(n)}) \\ \leq \exp \left\{ -\log \alpha_n - \alpha_n n^{3/8} + \frac{2n \alpha_n^2 s_{w,n}}{1 + s_{w,n}} \right\} \\ \leq \exp \left\{ -\log \alpha_n - \alpha_n n^{3/8} + 2n \alpha_n^2 s_{w,n} \right\} \\ = \frac{4n s_{w,n}}{n^{3/8}} \exp \left(-\frac{n^{3/4}}{8n s_{w,n}} \right). \end{aligned} \quad (68)$$

Since $ns_{w,n} = \Theta(\sqrt{n})$,

$$\frac{4ns_{w,n}}{n^{3/8}} = O(n^{1/8}), \quad \frac{n^{3/4}}{8ns_{w,n}} = \Theta(n^{1/4}), \quad (69)$$

and therefore

$$\mathbb{E}_{C_n} D\left(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) \rightarrow 0. \quad (70)$$

Thus, a sufficient resolvability condition is

$$\log M_n \geq nI_{w,n} + n^{3/8}. \quad (71)$$

Thus, in contrast to reliability, resolvability requires the codebook to contain sufficiently many codewords to approximate the Gaussian reference output.

Compatibility and Codebook-Size Selection. Reliability imposes an upper requirement on the admissible codebook size, whereas resolvability imposes a lower requirement. It remains to verify that the two conditions admit a common interval and to select an integer M_n within it. The Gaussian-input mutual information at Willie satisfies

$$\begin{aligned} nI_{w,n} &= n \log \left(1 + \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} \right) \\ &= \frac{|g_{aw}|^2}{N_w(P_d)} nQ_n + O(nQ_n^2) \\ &= (1 - \varepsilon)\sqrt{2\delta} \sqrt{n} + O(1). \end{aligned} \quad (72)$$

Combining this with the corresponding expansion of $nI_{b,n}$ gives

$$n(I_{b,n} - I_{w,n}) = (1 - \varepsilon)\sqrt{2\delta} \left[\frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} - 1 \right] \sqrt{n} + O(1). \quad (73)$$

Assume

$$\frac{|h_{ab}|^2}{N_b(P_d)} > \frac{|g_{aw}|^2}{N_w(P_d)}. \quad (74)$$

Then

$$n(I_{b,n} - I_{w,n}) = \Theta(\sqrt{n}), \quad (75)$$

with a strictly positive leading coefficient. Since $n^{3/8} = o(\sqrt{n})$, there exists n_0 such that, for all $n \geq n_0$,

$$n(I_{b,n} - I_{w,n}) \geq 2n^{3/8} + \log 2. \quad (76)$$

For such n , define

$$M_n \triangleq \left\lfloor \exp(nI_{b,n} - n^{3/8}) \right\rfloor. \quad (77)$$

Since $nI_{b,n} - n^{3/8} \rightarrow \infty$, for all sufficiently large n ,

$$nI_{b,n} - n^{3/8} - \log 2 \leq \log M_n \leq nI_{b,n} - n^{3/8}. \quad (78)$$

Consequently,

$$nI_{b,n} - \log M_n \geq n^{3/8}, \quad (79)$$

and

$$\begin{aligned} \log M_n - nI_{w,n} &\geq n(I_{b,n} - I_{w,n}) - n^{3/8} - \log 2 \\ &\geq n^{3/8}. \end{aligned} \quad (80)$$

Hence the selected M_n simultaneously satisfies the sufficient conditions for Bob reliability and Willie resolvability.

Moreover,

$$\begin{aligned} \log M_n &= nI_{b,n} - n^{3/8} + O(1) \\ &= (1 - \varepsilon)\sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + o(\sqrt{n}). \end{aligned} \quad (81)$$

Per-Codeword Power Constraint. Gaussian codeword generation satisfies the power constraint only probabilistically, so we must verify that all M_n codewords obey the prescribed per-codeword bound simultaneously. Define the event that all codewords in the random codebook satisfy the prescribed power constraint as

$$\mathcal{G}_n \triangleq \left\{ \frac{1}{n} \|X^n(m)\|^2 \leq P_a, \forall m \in \{1, \dots, M_n\} \right\}. \quad (82)$$

Its complement is therefore

$$\mathcal{G}_n^c = \left\{ \exists m \in \{1, \dots, M_n\} : \frac{1}{n} \|X^n(m)\|^2 > P_a \right\}. \quad (83)$$

Since $P_a > 0$ is fixed and $Q_n = \Theta(n^{-1/2})$,

$$\frac{P_a}{Q_n} \rightarrow \infty. \quad (84)$$

For any fixed codeword index m , the Gaussian construction gives

$$\frac{|X_i(m)|^2}{Q_n} \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(1), \quad i = 1, \dots, n,$$

and hence, for every $0 < t < 1$,

$$\mathbb{E} \exp\left(t \frac{|X_i(m)|^2}{Q_n}\right) = \frac{1}{1-t}.$$

Applying the Chernoff bound,

$$\begin{aligned}\Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] &= \Pr\left[\sum_{i=1}^n \frac{|X_i(m)|^2}{Q_n} > \frac{nP_a}{Q_n}\right] \\ &\leq \exp\left(-\frac{tnP_a}{Q_n}\right) \prod_{i=1}^n \mathbb{E} \exp\left(t \frac{|X_i(m)|^2}{Q_n}\right) \\ &= \exp\left\{-n \left[\frac{tP_a}{Q_n} + \log(1-t)\right]\right\}.\end{aligned}\quad (85)$$

Choosing $t = 1/2$ yields

$$\Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] \leq \exp\left\{-n \left[\frac{P_a}{2Q_n} - \log 2\right]\right\}.\quad (86)$$

Since $P_a/Q_n \rightarrow \infty$, for all sufficiently large n ,

$$\frac{P_a}{4Q_n} \geq \log 2,$$

and therefore

$$\Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] \leq \exp\left(-\frac{nP_a}{4Q_n}\right).\quad (87)$$

Using the definition of $\mathcal{G}_n^c$ and a union bound over all codewords,

$$\begin{aligned}\Pr(\mathcal{G}_n^c) &= \Pr\left[\bigcup_{m=1}^{M_n} \left\{\frac{1}{n}\|X^n(m)\|^2 > P_a\right\}\right] \\ &\leq \sum_{m=1}^{M_n} \Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] \\ &= M_n \Pr\left[\frac{1}{n}\|X^n(1)\|^2 > P_a\right] \\ &\leq \exp\left\{\log M_n - \frac{nP_a}{4Q_n}\right\}.\end{aligned}\quad (88)$$

For the previously selected codebook size,

$$\log M_n = O(\sqrt{n}),\quad (89)$$

whereas

$$Q_n = \Theta(n^{-1/2}) \implies \frac{nP_a}{Q_n} = \Theta(n^{3/2}).\quad (90)$$

Consequently,

$$\log M_n - \frac{nP_a}{4Q_n} = -\Theta(n^{3/2}),\quad (91)$$

and hence

$$\Pr(\mathcal{G}_n^c) \longrightarrow 0, \quad \Pr(\mathcal{G}_n) \longrightarrow 1.\quad (92)$$

Thus, with probability tending to one under the random-codebook ensemble, all codewords simultaneously satisfy

$$\frac{1}{n}\|X^n(m)\|^2 \leq P_a, \quad m = 1, \dots, M_n.\quad (93)$$

Actual Coverttness. We have shown that the random-codebook output approaches the Gaussian reference distribution. We now show that the resulting transmission also satisfies the original coverttness constraint at Willie. Let $p_0^{(n)}(z^n)$ denote the density of $P_0^{(n)}$. We first verify that the Gaussian reference distribution is exactly the ensemble-average output distribution induced by the random codebook. For every $z^n \in \mathbb{C}^n$,

$$\begin{aligned}\mathbb{E}_{\mathcal{C}_n} [p_{1,\mathcal{C}_n}^{(n)}(z^n)] &= \frac{1}{M_n} \sum_{m=1}^{M_n} \mathbb{E}_{X^n(m)} \left[\prod_{i=1}^n p_{Z|X}(z_i | X_i(m)) \right] \\ &= \prod_{i=1}^n \mathbb{E}_X [p_{Z|X}(z_i | X)] = \prod_{i=1}^n \bar{p}_{1,n}(z_i) = \bar{p}_{1,n}^{(n)}(z^n).\end{aligned}\quad (94)$$

Hence,

$$\mathbb{E}_{\mathcal{C}_n} [P_{1,\mathcal{C}_n}^{(n)}] = \bar{P}_{1,n}^{(n)}.\quad (95)$$

For every realization of the random codebook,

$$\begin{aligned}D(P_{1,\mathcal{C}_n}^{(n)} \| P_0^{(n)}) &= D(P_{1,\mathcal{C}_n}^{(n)} \| \bar{P}_{1,n}^{(n)}) \\ &\quad + \int_{\mathbb{C}^n} p_{1,\mathcal{C}_n}^{(n)}(z^n) \log \frac{\bar{p}_{1,n}^{(n)}(z^n)}{p_0^{(n)}(z^n)} dz^n.\end{aligned}\quad (96)$$

Taking expectation over $\mathcal{C}_n$ and using the preceding ensemble-average identity gives

$$\begin{aligned}\mathbb{E}_{\mathcal{C}_n} D(P_{1,\mathcal{C}_n}^{(n)} \| P_0^{(n)}) &= \mathbb{E}_{\mathcal{C}_n} D(P_{1,\mathcal{C}_n}^{(n)} \| \bar{P}_{1,n}^{(n)}) + D(\bar{P}_{1,n}^{(n)} \| P_0^{(n)}).\end{aligned}\quad (97)$$

Since

$$s_{w,n} = (1 - \varepsilon) \sqrt{\frac{2\delta}{n}},$$

the reference divergence is

$$D(\bar{P}_{1,n}^{(n)} \| P_0^{(n)}) = n[s_{w,n} - \log(1 + s_{w,n})]. \quad (98)$$

For every $x \geq 0$,

$$x - \log(1 + x) \leq \frac{x^2}{2}. \quad (99)$$

Indeed,

$$\frac{d}{dx} \left[\frac{x^2}{2} - x + \log(1 + x) \right] = \frac{x^2}{1 + x} \geq 0,$$

and the expression in brackets equals zero at $x = 0$. Therefore,

$$\begin{aligned} D(\bar{P}_{1,n}^{(n)} \| P_0^{(n)}) &\leq \frac{ns_{w,n}^2}{2} \\ &= (1 - \varepsilon)^2 \delta. \end{aligned} \quad (100)$$

Moreover, the preceding resolvability result gives

$$\mathbb{E}_{\mathcal{C}_n} D(P_{1,\mathcal{C}_n}^{(n)} \| \bar{P}_{1,n}^{(n)}) \rightarrow 0. \quad (101)$$

Since

$$\delta - (1 - \varepsilon)^2 \delta = \delta(2\varepsilon - \varepsilon^2) > 0, \quad (102)$$

there exists n_0 such that, for every $n \geq n_0$,

$$\mathbb{E}_{\mathcal{C}_n} D(P_{1,\mathcal{C}_n}^{(n)} \| \bar{P}_{1,n}^{(n)}) \leq \frac{\delta(2\varepsilon - \varepsilon^2)}{2}. \quad (103)$$

Consequently,

$$\begin{aligned} \mathbb{E}_{\mathcal{C}_n} D(P_{1,\mathcal{C}_n}^{(n)} \| P_0^{(n)}) &\leq (1 - \varepsilon)^2 \delta + \frac{\delta(2\varepsilon - \varepsilon^2)}{2} \\ &= \delta - \frac{\delta(2\varepsilon - \varepsilon^2)}{2} \\ &< \delta. \end{aligned} \quad (104)$$

Finally, since KL divergence is nonnegative, Markov's inequality gives, for all sufficiently large n ,

$$\begin{aligned} \Pr[D(P_{1,\mathcal{C}_n}^{(n)} \| P_0^{(n)}) > \delta] \\ &\leq \frac{\mathbb{E}_{\mathcal{C}_n} D(P_{1,\mathcal{C}_n}^{(n)} \| P_0^{(n)})}{\delta} \\ &\leq 1 - \frac{2\varepsilon - \varepsilon^2}{2}. \end{aligned} \quad (105)$$

Hence,

$$\Pr[D(P_{1,\mathcal{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta] \geq \frac{2\varepsilon - \varepsilon^2}{2} > 0. \quad (106)$$

Thus, for all sufficiently large n , the random-codebook ensemble contains codebook realizations satisfying the actual covertness constraint with probability bounded away from zero.

Deterministic Fixed-Codebook Extraction. We now show that at least one deterministic codebook satisfies reliability, covertness, and the power constraint simultaneously. For a realization $\mathcal{C}_n$ of the random codebook, let $P_e^{(n)}(\mathcal{C}_n)$ denote the average decoding error probability of its associated ML decoder. The preceding reliability analysis gives

$$\mathbb{E}_{\mathcal{C}_n} [P_e^{(n)}(\mathcal{C}_n)] \rightarrow 0. \quad (107)$$

Since $P_e^{(n)}(\mathcal{C}_n) \geq 0$, Markov's inequality yields

$$\begin{aligned} \Pr\left[P_e^{(n)}(\mathcal{C}_n) > \sqrt{\mathbb{E}_{\mathcal{C}_n} [P_e^{(n)}(\mathcal{C}_n)]}\right] \\ \leq \sqrt{\mathbb{E}_{\mathcal{C}_n} [P_e^{(n)}(\mathcal{C}_n)]} \rightarrow 0. \end{aligned} \quad (108)$$

The power-constraint analysis has established

$$\Pr(\mathcal{G}_n^c) \rightarrow 0, \quad (109)$$

where $\mathcal{G}_n$ is the event that every codeword satisfies the per-codeword power constraint. Moreover, the actual-covertness analysis gives, for all sufficiently large n ,

$$\Pr[D(P_{1,\mathcal{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta] \geq \frac{2\varepsilon - \varepsilon^2}{2} > 0. \quad (110)$$

Using the union bound on their complements,

$$\begin{aligned}
& \Pr \left[\mathcal{G}_n, P_e^{(n)}(\mathbf{C}_n) \leq \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]}, \right. \\
& \quad \left. D(P_{1,\mathbf{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta \right] \\
& \geq \Pr \left[D(P_{1,\mathbf{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta \right] \\
& \quad - \Pr(\mathcal{G}_n^c) \\
& \quad - \Pr \left[P_e^{(n)}(\mathbf{C}_n) > \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]} \right] \\
& \geq \frac{2\varepsilon - \varepsilon^2}{2} - \Pr(\mathcal{G}_n^c) - \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]}.
\end{aligned} \tag{111}$$

The last two terms vanish as $n \rightarrow \infty$, whereas $(2\varepsilon - \varepsilon^2)/2 > 0$. Hence, for all sufficiently large n ,

$$\Pr \left[\mathcal{G}_n, P_e^{(n)}(\mathbf{C}_n) \leq \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]}, D(P_{1,\mathbf{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta \right] > 0. \tag{112}$$

Therefore, for every sufficiently large n , there exists a deterministic realization

$$\mathcal{C}_n^* \tag{113}$$

of the random codebook such that

$$P_e^{(n)}(\mathcal{C}_n^*) \leq \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]} \rightarrow 0, \tag{114}$$

$$D(P_{1,\mathcal{C}_n^*}^{(n)} \| P_0^{(n)}) \leq \delta, \tag{115}$$

and

$$\frac{1}{n} \|x^n(m)\|^2 \leq P_a, \quad m = 1, \dots, M_n. \tag{116}$$

Thus $\{\mathcal{C}_n^*\}$ forms a sequence of deterministic fixed codebooks satisfying reliability, actual covertness, and the per-codeword power constraint simultaneously.

Since every realization of $\mathbf{C}_n$ contains exactly M_n codewords, the extraction does not alter the previously selected codebook size. Hence

$$\log M_n = (1 - \varepsilon)\sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + o(\sqrt{n}). \tag{117}$$

Achievable Square-Root Coefficient. Since the above bound holds for every fixed $0 < \varepsilon < 1$, letting ε approach zero gives the final achievable coefficient. For every fixed $0 < \varepsilon < 1$, the preceding construction yields a sequence of deterministic fixed codebooks $\{\mathcal{C}_n^*\}$ satisfying, for all sufficiently large n ,

$$P_e^{(n)}(\mathcal{C}_n^*) \rightarrow 0, \tag{118}$$

$$D(P_{1,\mathcal{C}_n^*}^{(n)} \| P_0^{(n)}) \leq \delta, \tag{119}$$

and

$$\frac{1}{n} \|x^n(m)\|^2 \leq P_a, \quad m = 1, \dots, M_n. \tag{120}$$

Moreover, deterministic-codebook extraction does not alter the previously selected codebook size, and hence

$$\begin{aligned}
\log M_n &= nI_{b,n} - n^{3/8} + O(1) \\
&= (1 - \varepsilon)\sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + o(\sqrt{n}).
\end{aligned} \tag{121}$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{\log M_n}{\sqrt{n}} = (1 - \varepsilon)\sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}. \tag{122}$$

Since $\mathcal{C}_n^*$ is an admissible deterministic codebook, the optimal codebook size satisfies

$$M_n^*(\delta) \geq M_n \tag{123}$$

for all sufficiently large n . Consequently,

$$\liminf_{n \rightarrow \infty} \frac{\log M_n^*(\delta)}{\sqrt{n}} \geq (1 - \varepsilon)\sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}. \tag{124}$$

The above inequality holds for every fixed $0 < \varepsilon < 1$. Taking the supremum over ε and using

$$\sup_{0 < \varepsilon < 1} (1 - \varepsilon) = 1$$

gives

$$\boxed{\liminf_{n \rightarrow \infty} \frac{\log M_n^*(\delta)}{\sqrt{n}} \geq \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}}. \tag{125}$$

□